# Land Science Research Methods

## Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

# Land Science Research Methods (A10466)

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| **Short Title:** | Land Science Research Methods |
| **Faculty** | Science and Computing |
| **Department** | Science |
| **Cluster** | Placement and Projects |
| **Author** | NMCCARTHY |
| **Credits** | 5 |
| **Level:** | Advanced |

## Description of Module / Aims

This module will introduce techniques that are necessary in a research environment, such as scientific reporting writing skills, presentation skills, utilising literature search strategies (database, journal and internet) and scientific referencing. This will be coupled with the planning and design of a scientific research project on their chosen topic in the context of a literature review. Where necesssary, due to seasonal factors involved in research in agriculture, horticulture or forestry, some students may also have to initiate setting up of their projects in this semester.

## Programmes

| | | |
|---|---|---|
| `SCIE-0068` | BSc (Hons) in Agricultural Science (`WD_SAGRI_B`) | 4 / 7 / M |
| `SCIE-0068` | BSc (Hons) in Land Management (in Agriculture) (`WD_SBSAG_B`) | 1 / 1 / M |
| `SCIE-0068` | BSc (Hons) in Land Management (in Forestry) (`WD_SBSFO_B`) | 1 / 1 / M |
| `SCIE-0068` | BSc (Hons) in Land Management (in Horticulture) (`WD_SBSHO_B`) | 1 / 1 / M |

## Indicative Content

- Experimental design.
- Scientific report-writing skills.
- Presentation skills
- Literature search strategies, including databases, literature, journal and internet resources.
- Scientific referencing according to Harvard referencing guidelines.

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Choose scientific literature relevant to a specialised research topic from a variety of sources.
2. Compile a literature review on a chosen research topic, using an appropriate referencing system.
3. Critique the literature sourced.
4. Design a specialised research project plan.

## Learning and Teaching Methods

- Lectures
- Library resources including databases (ScienceDirect, Web of Science, etc)
- Tutorials
- Self-directed study.

## Assessment Methods

| | Weighing | Outcomes Assessed |
|---|---|---|
| Continuous Assessment | 100% | |
| *Project* | 100% | 1,2,3,4 |

## Assessment Criteria

**<40%:** Very limited knowledge and understanding of the subject matter. Failure to meet the objectives of projects. Unable to carry out or submit independent study and research.

**40%-49%:** Ability to show a basic knowledge of research methodologies and basic competency in their use. Basic level of knowledge of chosen research topic.

**50%-59%:** As well as the above, the student will be able to demonstrate an understanding of some of the complexities of research methodologies and a good competency in their use. Display an adequate level of knowledge of chosen research topic supplemented by an ability to present this knowledge in a clear and concise manner.

**60%-69%:** In addition to the above, the student will demonstrate a more detailed knowledge of research methodologies and a high level of competence in their use. Display a high level of knowledge of chosen research topic, supplemented by an ability to present this knowledge to a high technical standard.

**70%-100%:** All of the above to an excellent level. In addition, the student will demonstrate an advanced knowledge of research methodologies, as well as excellent and proficient competence in their use. Display a thorough level of knowledge of chosen research topic, supplemented by an ability to critically evaluate and discuss the topic in relation to a wider scientific context. The above advanced knowledge of the topic will be supported by superior reporting and presentation skills.

## Learning Modes

| Learning Mode | F/T Hours | P/T Hours |
|---|---|---|
| Lecture | 36 | |
| Independent Learning | 99 | |

## Essential Material

- "Online resources & databases." www.google.com
- "ScienceDirect." www.sciencedirect.com
- "Web of Science." wos.heanet.ie
- Bell, J. and S. walters. _Doing Your Research project: A guide for first-time researchers_. 6th ed.. UK: Open University Press, 2014.
- Various, A. _Research Papers and References_. World: Various, Varied.

## Requested Resources

- Lecture Room: Loose Seated
- Room Type: Computer Lab